A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-2800905})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 27 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \mathrm{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 906 & 483 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,906 + \mathbf{Z}\,(483 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 2800905 = 0$, of discriminant -11203620. Class group invariants $(10 \quad 10 \quad 10)$, class number 1000; the three stored generator ideals (HNF) are $\begin{pmatrix} 426 & 51 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1255 & 755 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 13 & 7 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 1027y^3 + (8s - 33869)\,y^2 + (340s - 648684)\,y + (2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 2053*y^8 - 65684*y^7 - 174901*y^6 + 64171174*y^5 + 26
65457137*y^4 + 66051314232*y^3 + 1091726153376*y^2 + 9461326049280*y
+ 31436965969920
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right) s, \qquad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $13070819502 = 2 \cdot 3^4 \cdot 13 \cdot 179 \cdot 34673$:

$$\begin{pmatrix}
484104426 & 375659442 & 231314970 & 20784355 & 196407038 & 443935002 & 450125258 & 456136876 & 450734961 & 164151061 \\
0 & 3 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 3 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 130 factors with signed exponents of absolute value up to 61432318; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 1027y^3 + (8s - 33869)\, y^2 + (340s - 648684)\, y + (2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 2053*y^8 - 65684*y^7 - 174901*y^6 + 64171174*y^5 + 26
65457137*y^4 + 66051314232*y^3 + 1091726153376*y^2 + 9461326049280*y
+ 31436965969920
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + (\frac{16828}{2498705294406001})\, s, \qquad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $10587476640 = 2^5 \cdot 3 \cdot 5 \cdot 13 \cdot 1696711$:

$$\begin{pmatrix}
661717290 & 304571440 & 481114626 & 490455262 & 495451507 & 319162329 & 161602581 & 601870900 & 618810446 & 621630709 \\
0 & 2 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 128 factors with signed exponents of absolute value up to 27582645; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 3 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 1027y^3 + (8s - 33869)\,y^2 + (340s - 648684)\,y + (2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 2053*y^8 - 65684*y^7 - 174901*y^6 + 64171174*y^5 + 26
65457137*y^4 + 66051314232*y^3 + 1091726153376*y^2 + 9461326049280*y
+ 31436965969920
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + (\frac{191}{137858491849})\,s, \qquad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2099627642060 = 2^2 \cdot 5 \cdot 13 \cdot 2789 \cdot 2895479$:

$$\begin{pmatrix}
1049813821030 & 734127432014 & 67173029314 & 959389486830 & 57053121859 & 57785365490 & 392561043258 & 895986675712 & 841915140590 & 740512538666 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 133 factors with signed exponents of absolute value up to 63178993; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 680y^3 + 8sy^2 - 115836y - 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 1360*y^8 + 230728*y^6 + 21720960*y^4 + 2662503696*y^2 + 16133
2128000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right) s, \qquad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $412192614796 = 2^2 \cdot 107 \cdot 963066857$:

$$\begin{pmatrix} 206096307398 & 164520907200 & 126567181928 & 49396419060 & 40840969302 & 9830235395 & 196983202531 & 175469218589 & 11327597003 & 146052724215 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 428347; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 680y^3 + 8sy^2 - 115836y - 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 1360*y^8 + 230728*y^6 + 21720960*y^4 + 2662503696*y^2 + 16133
2128000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + (\frac{16828}{2498705294406001})\, s, \qquad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $247528120320 = 2^{13} \cdot 3^5 \cdot 5 \cdot 13 \cdot 1913$:

$$\begin{pmatrix}
2984280 & 1860624 & 2148252 & 1468916 & 1851016 & 979330 & 635075 & 2060362 & 378920 & 2299871 \\
0 & 24 & 18 & 0 & 9 & 0 & 2 & 5 & 10 & 7 \\
0 & 0 & 6 & 4 & 5 & 2 & 4 & 1 & 0 & 2 \\
0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 6 & 0 & 2 & 2 & 4 & 1 \\
0 & 0 & 0 & 0 & 0 & 12 & 0 & 10 & 0 & 6 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 159 factors with signed exponents of absolute value up to 18128; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 680y^3 + 8sy^2 - 115836y - 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 1360*y^8 + 230728*y^6 + 21720960*y^4 + 2662503696*y^2 + 16133
2128000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + (\frac{191}{137858491849})\, s, \qquad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1633673869360 = 2^4 \cdot 5 \cdot 13 \cdot 67^2 \cdot 349931$:

$$\begin{pmatrix} 3047899010 & 641341554 & 683872202 & 2867786190 & 1993074766 & 2128589928 & 2190478923 & 1411705703 & 1734308691 & 2031171313 \\ 0 & 134 & 123 & 10 & 132 & 116 & 0 & 0 & 50 & 83 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 164 factors with signed exponents of absolute value up to 1065502; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 123)\, y^3 + (10s + 24508)\, y^2 + (14s - 189214)\, y + (-240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 245*y^8 + 49262*y^7 + 2388590*y^6 - 60834832*y^5 + 84
8020060*y^4 - 7248384008*y^3 + 43341537640*y^2 - 176590228512*y + 33
5141073216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + (\frac{403}{16956588762216})\, s, \qquad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $244984828968 = 2^3 \cdot 3^3 \cdot 11^3 \cdot 113 \cdot 7541$:

$$\begin{pmatrix} 1237297116 & 1107664602 & 373802134 & 673928382 & 599870118 & 19670863 & 259627519 & 441083314 & 19113414 & 997088270 \\ 0 & 66 & 21 & 51 & 33 & 42 & 15 & 33 & 45 & 49 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 2 & 2 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 113 factors with signed exponents of absolute value up to 30412; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 123)\, y^3 + (10s + 24508)\, y^2 + (14s - 189214)\, y + (-240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 245*y^8 + 49262*y^7 + 2388590*y^6 - 60834832*y^5 + 84
8020060*y^4 - 7248384008*y^3 + 43341537640*y^2 - 176590228512*y + 33
5141073216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right) s, \qquad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $7544716245 = 3^6 \cdot 5 \cdot 11 \cdot 188171$:

$$\begin{pmatrix} 93144645 & 77474589 & 31336536 & 33924207 & 1400637 & 39112834 & 59202031 & 87371245 & 8148687 & 9424261 \\ 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 3 & 0 & 1 & 2 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 3 & 1 & 2 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 116 factors with signed exponents of absolute value up to 113726; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 123)\, y^3 + (10s + 24508)\, y^2 + (14s - 189214)\, y + (-240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 245*y^8 + 49262*y^7 + 2388590*y^6 - 60834832*y^5 + 84
8020060*y^4 - 7248384008*y^3 + 43341537640*y^2 - 176590228512*y + 33
5141073216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right) s, \qquad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $583425695160 = 2^3 \cdot 3^4 \cdot 5 \cdot 7^2 \cdot 11^4 \cdot 251$:

$$\begin{pmatrix} 579810 & 390654 & 439978 & 400906 & 157986 & 129878 & 22438 & 118162 & 331209 & 183238 \\ 0 & 66 & 0 & 0 & 27 & 30 & 30 & 45 & 42 & 7 \\ 0 & 0 & 154 & 121 & 33 & 121 & 105 & 6 & 6 & 137 \\ 0 & 0 & 0 & 33 & 6 & 8 & 4 & 3 & 12 & 31 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 106 factors with signed exponents of absolute value up to 8752; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s - 62)\, y^3 + (-7s - 11955)\, y^2 + (6s - 75342)\, y + (1128s - 626400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 120*y^8 - 23662*y^7 + 2701885*y^6 - 38681682*y^5 + 32
5625238*y^4 + 7962666480*y^3 - 23477418216*y^2 + 132301507680*y + 39
56203667520
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + (\frac{403}{16956588762216})\, s, \qquad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $340301732544 = 2^6 \cdot 3^3 \cdot 241 \cdot 817153$:

$$\begin{pmatrix}
2363206476 & 1646932974 & 1685191470 & 395697078 & 1588385436 & 1086286908 & 793139898 & 255832331 & 1655470358 & 1595617503 \\
0 & 3 & 1 & 2 & 2 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 6 & 4 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 124 factors with signed exponents of absolute value up to 1003112; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s - 62)\,y^3 + (-7s - 11955)\,y^2 + (6s - 75342)\,y + (1128s - 626400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 120*y^8 - 23662*y^7 + 2701885*y^6 - 38681682*y^5 + 32
5625238*y^4 + 7962666480*y^3 - 23477418216*y^2 + 132301507680*y + 39
56203667520
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right) s, \qquad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1655028426480 = 2^4 \cdot 3 \cdot 5 \cdot 11^2 \cdot 13 \cdot 4383949$:

$$\begin{pmatrix}
18807141210 & 1149115242 & 9794350704 & 3804712500 & 13427970918 & 4275596576 & 15626843533 & 9182443815 & 15547433924 & 9395912319 \\
0 & 11 & 2 & 0 & 1 & 0 & 5 & 5 & 9 & 1 \\
0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 124 factors with signed exponents of absolute value up to 919270; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s - 62)\, y^3 + (-7s - 11955)\, y^2 + (6s - 75342)\, y + (1128s - 626400)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 120*y^8 - 23662*y^7 + 2701885*y^6 - 38681682*y^5 + 32
5625238*y^4 + 7962666480*y^3 - 23477418216*y^2 + 132301507680*y + 39
56203667520
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right) s, \qquad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $97407492600 = 2^3 \cdot 3 \cdot 5^2 \cdot 11^2 \cdot 1341701$:

$$\begin{pmatrix}
4870374630 & 1915556880 & 3019615650 & 1976487432 & 1779492132 & 3279218472 & 4006393506 & 559354305 & 543373656 & 3289497063 \\
0 & 10 & 4 & 6 & 4 & 2 & 0 & 7 & 7 & 4 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 130 factors with signed exponents of absolute value up to 1242228; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 17$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 4 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 121)\, y^3 + (16s + 3420)\, y^2 + (-180s - 305964)\, y + (-432s + 2001024)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 233*y^8 + 7566*y^7 + 2183098*y^6 - 84618768*y^5 + 179
9091024*y^4 - 16290272448*y^3 + 159330584736*y^2 - 788885868672*y +
4526813143296
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + (\frac{403}{16956588762216})\, s, \qquad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $233867734740 = 2^2 \cdot 3^2 \cdot 5 \cdot 23 \cdot 761 \cdot 74231$:

$$\begin{pmatrix} 38977955790 & 22161669858 & 1286145072 & 18112470847 & 32103976277 & 4808421144 & 27405158875 & 17252601535 & 14450236729 & 30371689007 \\ 0 & 6 & 0 & 2 & 3 & 5 & 0 & 3 & 3 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 129 factors with signed exponents of absolute value up to 38956; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 121)\, y^3 + (16s + 3420)\, y^2 + (-180s - 305964)\, y + (-432s + 2001024)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 233*y^8 + 7566*y^7 + 2183098*y^6 - 84618768*y^5 + 179
9091024*y^4 - 16290272448*y^3 + 159330584736*y^2 - 788885868672*y +
4526813143296
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + (\frac{16828}{2498705294406001})\, s, \qquad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $22894558920 = 2^3 \cdot 3^2 \cdot 5 \cdot 17 \cdot 607 \cdot 6163$:

$$\begin{pmatrix}
1907879910 & 1327076322 & 823966894 & 644951425 & 329851431 & 1220174512 & 1798843377 & 1470964105 & 717074021 & 994683121 \\
0 & 6 & 4 & 5 & 1 & 0 & 5 & 0 & 1 & 0 \\
0 & 0 & 2 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 124 factors with signed exponents of absolute value up to 243599; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 4 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + (-s - 121)\, y^3 + (16s + 3420)\, y^2 + (-180s - 305964)\, y + (-432s + 2001024)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 - 233*y^8 + 7566*y^7 + 2183098*y^6 - 84618768*y^5 + 179
9091024*y^4 - 16290272448*y^3 + 159330584736*y^2 - 788885868672*y +
4526813143296
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + (\frac{191}{137858491849})\, s, \qquad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2154434142720 = 2^9 \cdot 3 \cdot 5 \cdot 17 \cdot 1861 \cdot 8867$:

$$\begin{pmatrix} 16831516740 & 15024422532 & 7067870470 & 12361579048 & 12286250382 & 13379749198 & 9207865976 & 15603721634 & 4050201654 & 7121128570 \\ 0 & 4 & 2 & 2 & 2 & 0 & 2 & 1 & 3 & 1 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 129 factors with signed exponents of absolute value up to 100605; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 26y^3 - 12sy^2 + 483525y + 2220s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 52*y^8 + 967726*y^6 + 428473620*y^4 + 84564207225*y^2 + 13803
980202000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + (\frac{403}{16956588762216})\, s, \qquad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $178513655400 = 2^3 \cdot 3^2 \cdot 5^2 \cdot 163 \cdot 608431$:

$$\begin{pmatrix}
2975227590 & 186859530 & 2156152170 & 2723959661 & 1624111463 & 1550589379 & 894969644 & 1402533441 & 1806190805 & 378457270 \\
0 & 30 & 0 & 29 & 23 & 0 & 17 & 13 & 14 & 26 \\
0 & 0 & 2 & 1 & 1 & 0 & 1 & 0 & 1 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 157 factors with signed exponents of absolute value up to 124585997; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 26y^3 - 12sy^2 + 483525y + 2220s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 52*y^8 + 967726*y^6 + 428473620*y^4 + 84564207225*y^2 + 13803
980202000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + (\frac{16828}{2498705294406001})\, s, \qquad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $372996494700 = 2^2 \cdot 3 \cdot 5^2 \cdot 23 \cdot 179 \cdot 301997$:

$$\begin{pmatrix}
37299649470 & 12584814375 & 4863013453 & 12888650423 & 33161757620 & 28979049959 & 20349123815 & 3239344032 & 6671544396 & 21285579387 \\
0 & 5 & 4 & 0 & 4 & 3 & 3 & 4 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 147592620; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 26y^3 - 12sy^2 + 483525y + 2220s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 52*y^8 + 967726*y^6 + 428473620*y^4 + 84564207225*y^2 + 13803
980202000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 1 \quad -1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + (\frac{191}{137858491849})\, s, \qquad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $55781154432 = 2^7 \cdot 3^2 \cdot 23 \cdot 2105267$:

$$\begin{pmatrix} 290526846 & 233846454 & 14543714 & 211131226 & 203783830 & 231580250 & 96562340 & 91623721 & 172620616 & 111229417 \\ 0 & 6 & 0 & 4 & 1 & 0 & 2 & 0 & 4 & 3 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 155 factors with signed exponents of absolute value up to 8392999; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 19: character $2a$, column e_1

Prescribed character vector $(2 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 1027y^3 + (-8s - 33869)\,y^2 + (-340s - 648684)\,y + (-2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 2053*y^8 - 65684*y^7 - 174901*y^6 + 64171174*y^5 + 26
65457137*y^4 + 66051314232*y^3 + 1091726153376*y^2 + 9461326049280*y
+ 31436965969920
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad -1 \quad -1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right) s, \qquad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $13070819502 = 2 \cdot 3^4 \cdot 13 \cdot 179 \cdot 34673$:

$$\begin{pmatrix} 484104426 & 375659442 & 231314970 & 20784355 & 196407038 & 443935002 & 450125258 & 456136876 & 450734961 & 164151061 \\ 0 & 3 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 3 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 128 factors with signed exponents of absolute value up to 150967808; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 20: character $2a$, column e_2

Prescribed character vector $(2 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 1027y^3 + (-8s - 33869)\, y^2 + (-340s - 648684)\, y + (-2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 2053*y^8 - 65684*y^7 - 174901*y^6 + 64171174*y^5 + 26
65457137*y^4 + 66051314232*y^3 + 1091726153376*y^2 + 9461326049280*y
+ 31436965969920
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad -1 \quad -1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right) s, \qquad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $379155860616 = 2^3 \cdot 3 \cdot 13 \cdot 499 \cdot 2435357$:

$$\begin{pmatrix}
94788965154 & 64360777590 & 1897812564 & 3211914925 & 45867235664 & 8730312770 & 91438839032 & 77122522922 & 12217977932 & 54528951723 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 2 & 1 & 1 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 134 factors with signed exponents of absolute value up to 266216466; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 2 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 21: character $2a$, column e_3

Prescribed character vector $(2 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 - 1027y^3 + (-8s - 33869)\, y^2 + (-340s - 648684)\, y + (-2688s - 3346560)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 2053*y^8 - 65684*y^7 - 174901*y^6 + 64171174*y^5 + 26
65457137*y^4 + 66051314232*y^3 + 1091726153376*y^2 + 9461326049280*y
+ 31436965969920
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 0 \quad -1 \quad 0 \quad 0 \quad -1 \quad -1 \quad -1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right) s, \qquad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $158406917760 = 2^7 \cdot 3^6 \cdot 5 \cdot 7^2 \cdot 13^2 \cdot 41$:

$$\begin{pmatrix} 223860 & 119406 & 179382 & 86262 & 70458 & 48720 & 98598 & 193694 & 138302 & 143952 \\ 0 & 21 & 12 & 12 & 0 & 3 & 0 & 15 & 6 & 17 \\ 0 & 0 & 78 & 24 & 60 & 66 & 48 & 18 & 54 & 58 \\ 0 & 0 & 0 & 3 & 2 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 128 factors with signed exponents of absolute value up to 363594176; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 22: character $2b$, column e_1

Prescribed character vector $(0 \quad 2 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 680y^3 - 8sy^2 - 115836y + 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 1360*y^8 + 230728*y^6 + 21720960*y^4 + 2662503696*y^2 + 16133
2128000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right) s, \qquad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5293226700 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 557 \cdot 10559$:

$$\begin{pmatrix} 176440890 & 5809020 & 87211030 & 83818622 & 7566778 & 30593077 & 86521691 & 5971211 & 18737894 & 35710437 \\ 0 & 15 & 14 & 8 & 8 & 0 & 11 & 0 & 0 & 14 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 158 factors with signed exponents of absolute value up to 1397128; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 23: character $2b$, column e_2

Prescribed character vector $(0 \quad 2 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 680y^3 - 8sy^2 - 115836y + 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 1360*y^8 + 230728*y^6 + 21720960*y^4 + 2662503696*y^2 + 16133
2128000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right) s, \qquad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $213512823864 = 2^3 \cdot 3^2 \cdot 107 \cdot 509 \cdot 54449$:

$$\begin{pmatrix} 17792735322 & 12119088490 & 6549266688 & 12824107860 & 9112035044 & 11884733755 & 17024466190 & 7019531977 & 4967762761 & 7957321765 \\ 0 & 2 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 161 factors with signed exponents of absolute value up to 391; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 24: character $2b$, column e_3

Prescribed character vector $(0 \quad 2 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 680y^3 - 8sy^2 - 115836y + 240s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 1360*y^8 + 230728*y^6 + 21720960*y^4 + 2662503696*y^2 + 16133
2128000
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 1 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right) s, \qquad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $85548070400 = 2^9 \cdot 5^2 \cdot 13^2 \cdot 71 \cdot 557$:

$$\begin{pmatrix}
20564440 & 152776 & 4253744 & 9992654 & 11512468 & 7676454 & 20314856 & 9135484 & 18860211 & 1563759 \\
0 & 52 & 28 & 33 & 34 & 22 & 34 & 8 & 42 & 41 \\
0 & 0 & 10 & 0 & 0 & 9 & 9 & 2 & 7 & 9 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 163 factors with signed exponents of absolute value up to 4221650; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 2; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 25: character $2c$, column e_1

Prescribed character vector $(0 \quad 0 \quad 2)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s - 123)\, y^3 + (-10s + 24508)\, y^2 + (-14s - 189214)\, y + (240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 245*y^8 + 49262*y^7 + 2388590*y^6 - 60834832*y^5 + 84
8020060*y^4 - 7248384008*y^3 + 43341537640*y^2 - 176590228512*y + 33
5141073216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{4703}{628021806008} + \left(\frac{403}{16956588762216}\right) s, \qquad J = \begin{pmatrix} 906 & 483 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 906$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $102545175744 = 2^6 \cdot 3^8 \cdot 11 \cdot 149^2$:

$$\begin{pmatrix}
59004 & 17028 & 20136 & 3336 & 57378 & 38613 & 15177 & 35244 & 15977 & 27605 \\
0 & 36 & 18 & 6 & 30 & 18 & 22 & 11 & 0 & 0 \\
0 & 0 & 894 & 66 & 477 & 798 & 191 & 310 & 872 & 844 \\
0 & 0 & 0 & 6 & 3 & 0 & 1 & 2 & 0 & 2 \\
0 & 0 & 0 & 0 & 3 & 0 & 1 & 0 & 0 & 2 \\
0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 0 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 108 factors with signed exponents of absolute value up to 57768; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 26: character $2c$, column e_2

Prescribed character vector $(0 \quad 0 \quad 2)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s - 123)\, y^3 + (-10s + 24508)\, y^2 + (-14s - 189214)\, y + (240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 245*y^8 + 49262*y^7 + 2388590*y^6 - 60834832*y^5 + 84
8020060*y^4 - 7248384008*y^3 + 43341537640*y^2 - 176590228512*y + 33
5141073216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41298191}{2498705294406001} + \left(\frac{16828}{2498705294406001}\right) s, \qquad J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1201$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $357730716090 = 2 \cdot 3 \cdot 5 \cdot 11 \cdot 71 \cdot 151 \cdot 101113$:

$$\begin{pmatrix}
357730716090 & 279888645419 & 239627999627 & 114743485313 & 172262182433 & 65940137727 & 4728492971 & 219291936158 & 99122809393 & 154733809858 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 117 factors with signed exponents of absolute value up to 49107; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 27: character $2c$, column e_3

Prescribed character vector $(0\quad 0\quad 2)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (s - 123)\, y^3 + (-10s + 24508)\, y^2 + (-14s - 189214)\, y + (240s + 416904)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 245*y^8 + 49262*y^7 + 2388590*y^6 - 60834832*y^5 + 84
8020060*y^4 - 7248384008*y^3 + 43341537640*y^2 - 176590228512*y + 33
5141073216
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0\quad -1\quad 0\quad 0\quad 0\quad 0\quad 1\quad 0\quad 0\quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{188888}{137858491849} + \left(\frac{191}{137858491849}\right) s, \qquad J = \begin{pmatrix} 169 & 59 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 169$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2474463940740 = 2^2 \cdot 3^5 \cdot 5 \cdot 11 \cdot 113 \cdot 157 \cdot 2609$:

$$\begin{pmatrix}
45823406310 & 20684903568 & 25290118320 & 35583112656 & 18983639054 & 44214279718 & 26842305973 & 17060040143 & 7532526746 & 37334140205 \\
0 & 6 & 3 & 3 & 0 & 3 & 3 & 3 & 3 & 1 \\
0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
0 & 0 & 0 & 3 & 0 & 2 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 106 factors with signed exponents of absolute value up to 11365; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0\quad 4\quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.